

Total No. of Questions : 4]

SEAT No. :

PB-390

[Total No. of Pages : 2

[6270] 202

**B.E. (Artificial Intelligence and Data Science) (Insem)**

**REINFORCEMENT LEARNING**

**(2019 Pattern) (Semester - VIII) (417533(D)) (Elective - VI)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) *Answer Q1 or Q2, Q3 or Q4.*
- 2) *Figures to the right indicate full marks.*
- 3) *Neat diagrams must be drawn wherever necessary.*
- 4) *Assume suitable data if necessary.*

**Q1) a)** What is reinforcement learning? Compare RL with other ML techniques. **[6]**

b) How has reinforcement learning evolved over time, from its early theoretical roots to practical applications in various domains? **[5]**

c) Explain limitation of reinforcement learning. **[4]**

OR

**Q2) a)** What is reinforcement learning? Explain one practical example. **[6]**

b) Explain how reinforcement learning influenced robotics and autonomous systems development? **[5]**

c) Explain various practical applications of reinforcement learning. **[4]**

**Q3) a)** What are the key components of a Markov decision process (MDP), and how do they formalize a reinforcement learning problem? **[6]**

b) Discuss the difference between policy evaluation and policy improvement in the context of Markov decision process (MDP). **[5]**

c) Explain the concept of infinite horizons in reinforcement learning. **[4]**

OR

- Q4)** a) Describe the Bellman equation for both the state-value function and the action-value function in MDPs, and discuss their significance in reinforcement learning algorithms. [6]
- b) Explain sequence of rewards assumption in reinforcement learning. [5]
- c) Discuss the Markov Properties. [4]

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